

Initiating Search

November 4, 2025, 9:27 PM

• Search:

CAS Lexicon Search:

Reductive amination - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (6,798)	 References	View Results
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Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2024 to 2025	

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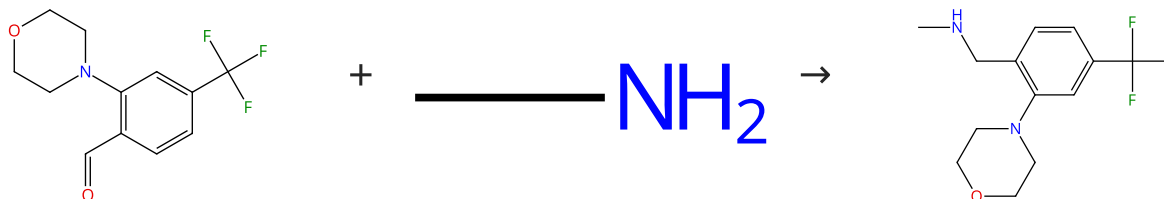
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (11)

 Suppliers (115)

 Suppliers (2)

31-614-CAS-40490477

Steps: 1 Yield: 100%

Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution

By: Jesani, Mehul H.; et al

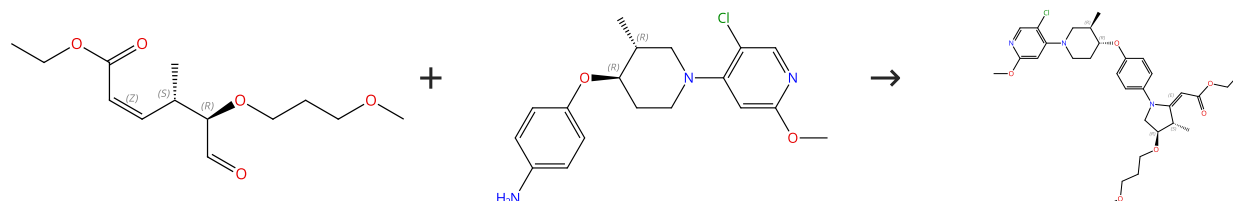
Angewandte Chemie, International Edition (2024), 63(24), e202403477.

- 1.1 **Solvents:** Ethanol, Methanol; 1.5 h, rt; rt → 0 °C
- 1.2 **Reagents:** Sodium borohydride; 0 °C; 0 °C → rt; 15 h, rt
- 1.3 **Reagents:** Sodium hydroxide
Solvents: Water

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown
Double bond geometry shown

Absolute stereochemistry shown

Absolute stereochemistry shown
Double bond geometry shown

 Supplier (1)

31-614-CAS-38944390

Steps: 1 Yield: 100%

Evolution of the Synthetic Process to an Advanced GPR40 Agonist

By: Hou, Xiaoping; et al

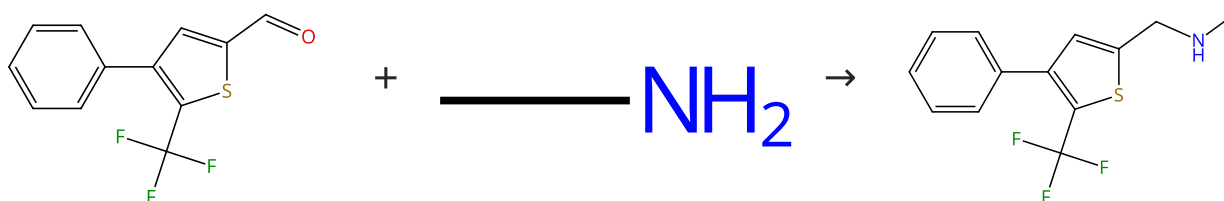
Organic Process Research & Development (2024), 28(1), 310-324.

- 1.1 **Reagents:** Sodium triacetoxyborohydride
Solvents: Dichloromethane; 1 h, 21 - 23 °C; 50 min, rt; rt → 0 °C
- 1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; 15 min, rt

Experimental Protocols

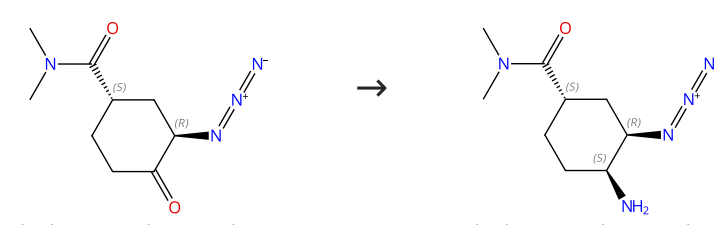
Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

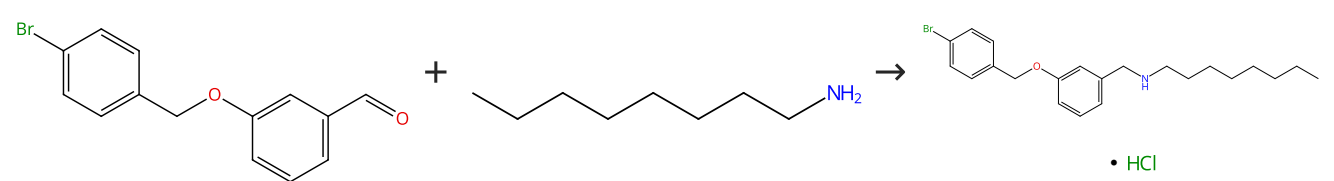

 Suppliers (38)

 Suppliers (115)

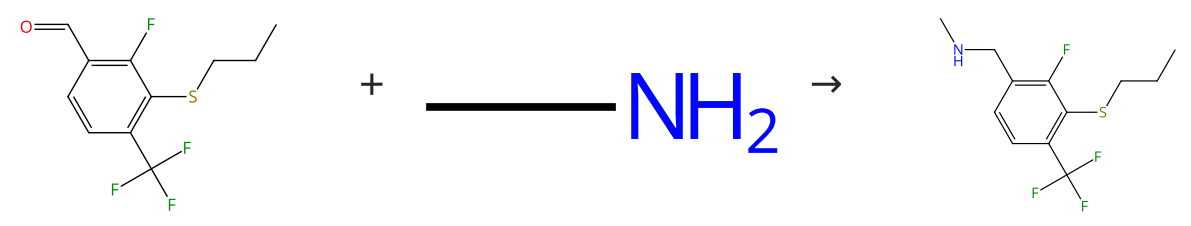
31-614-CAS-40490510 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol, Methanol; 16 h, rt; rt → 0 °C 1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 2 h, rt 1.3 Reagents: Sodium hydroxide Solvents: Water Experimental Protocols	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution By: Jesani, Mehul H.; et al Angewandte Chemie, International Edition (2024), 63(24), e202403477.
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Scheme 4 (1 Reaction)  Absolute stereochemistry shown, Rotation (+) Absolute stereochemistry shown, Rotation (-)	Steps: 1 Yield: 100%
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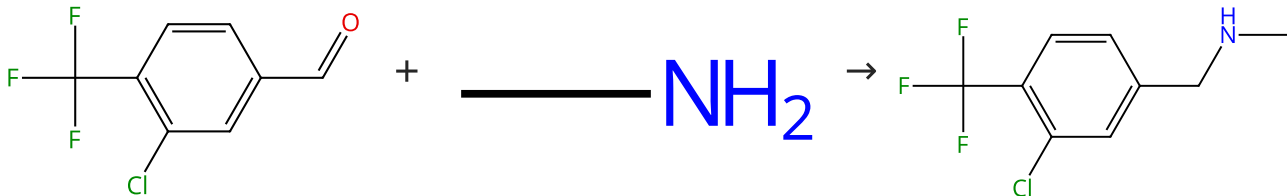
31-614-CAS-40647308 Steps: 1 Yield: 100% 1.1 Reagents: Isopropylamine, Hydrochloric acid, Pyridoxal 5'-phosphate monohydrate, β-Alanine:pyruvate aminotransferase Solvents: Dimethyl sulfoxide, Water; pH 7.5, 30 °C; 2 h, pH 7.4 - 7.6, 30 °C; 24 h Experimental Protocols	A New Method to Large-Scalable Preparation of the Key Chiral Cyclohexane cis-Diamine Intermediate for Edoxaban By: Yuan, Dingding; et al Organic Process Research & Development (2024), 28(7), 2623-2634.
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Scheme 5 (1 Reaction)  • HCl  Suppliers (39)  Suppliers (79)	Steps: 1 Yield: 100%
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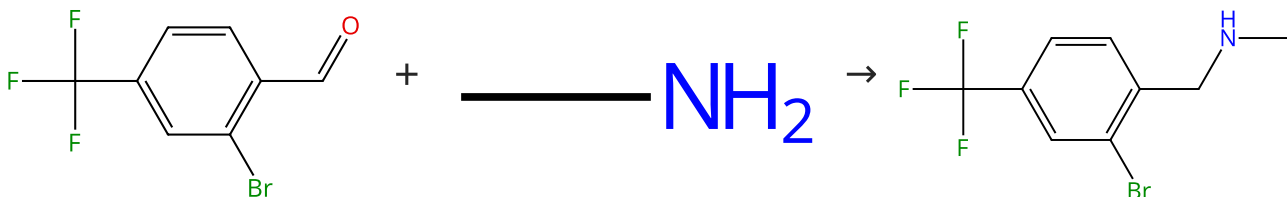
31-614-CAS-39391464 Steps: 1 Yield: 100% 1.1 Reagents: Sodium triacetoxyborohydride Solvents: Tetrahydrofuran; 12 h, rt 1.2 Reagents: Sodium bicarbonate Solvents: Water 1.3 Reagents: Hydrochloric acid Solvents: Diethyl ether Experimental Protocols	Synthesis of novel benzylamine antimycotics and evaluation of their antimycotic potency By: Klimt, Monika; et al Archiv der Pharmazie (Weinheim, Germany) (2024), 357(5), 2300381.
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Scheme 6 (1 Reaction)   Suppliers (115)	Steps: 1 Yield: 100%
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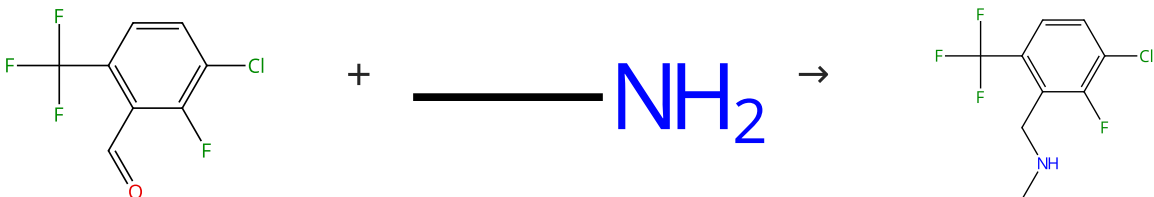
31-614-CAS-40490535 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol, Methanol; 1.5 h, rt; rt → 0 °C 1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 30 min, rt 1.3 Reagents: Sodium hydroxide Solvents: Water Experimental Protocols	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution By: Jesani, Mehul H.; et al Angewandte Chemie, International Edition (2024), 63(24), e202403477.
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Scheme 7 (1 Reaction)   Suppliers (59)  Suppliers (115)  Suppliers (8)	Steps: 1 Yield: 100%
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31-614-CAS-40490484 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol, Methanol; 1.5 h, rt; rt → 0 °C 1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 3.5 h, rt 1.3 Reagents: Sodium hydroxide Solvents: Water Experimental Protocols	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution By: Jesani, Mehul H.; et al Angewandte Chemie, International Edition (2024), 63(24), e202403477.
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Scheme 8 (1 Reaction)   Suppliers (63)  Suppliers (115)  Suppliers (18)	Steps: 1 Yield: 100%
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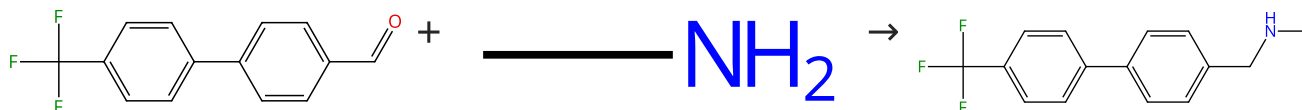
31-614-CAS-40490483 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol, Methanol; 1.5 h, rt; rt → 0 °C 1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 25 h, rt 1.3 Reagents: Sodium hydroxide Solvents: Water Experimental Protocols	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution By: Jesani, Mehul H.; et al Angewandte Chemie, International Edition (2024), 63(24), e202403477.
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Scheme 9 (1 Reaction)   Suppliers (51)  Suppliers (115)	Steps: 1 Yield: 100%
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31-614-CAS-40490487	Steps: 1 Yield: 100%	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution
1.1 Solvents: Ethanol, Methanol; 1.5 h, rt; rt → 0 °C		By: Jesani, Mehul H.; et al
1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 1.5 h, rt		Angewandte Chemie, International Edition (2024), 63(24), e202403477.
1.3 Reagents: Sodium hydroxide Solvents: Water		
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

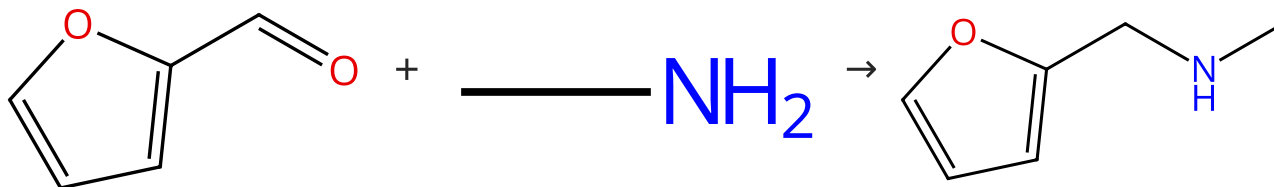
Suppliers (115)

Suppliers (9)

31-614-CAS-40490497	Steps: 1 Yield: 100%	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution
1.1 Solvents: Ethanol, Methanol, Tetrahydrofuran; 1.5 h, rt; rt → 0 °C		By: Jesani, Mehul H.; et al
1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 15 h, rt		Angewandte Chemie, International Edition (2024), 63(24), e202403477.
1.3 Reagents: Sodium hydroxide Solvents: Water		
Experimental Protocols		

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (81)

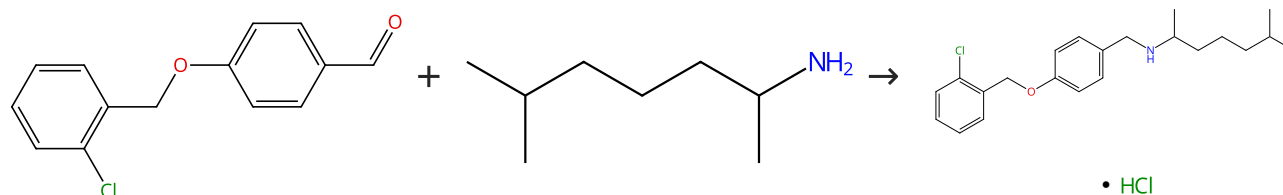
Suppliers (115)

Suppliers (62)

31-614-CAS-40172273	Steps: 1 Yield: 100%	Chemoenzymatic synthesis of amino-esters as precursors of ammonium salt-based surfactants from 5-hydroxymethylfurfural (HMF)
1.1 Reagents: Hydrogen Catalysts: Carbon, Cobalt Solvents: Ethanol; 6 h, 4 bar, 60 °C		By: Moriana Herraiz, Carlos; et al
Experimental Protocols		Green Chemistry (2024), 26(16), 9118-9131.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



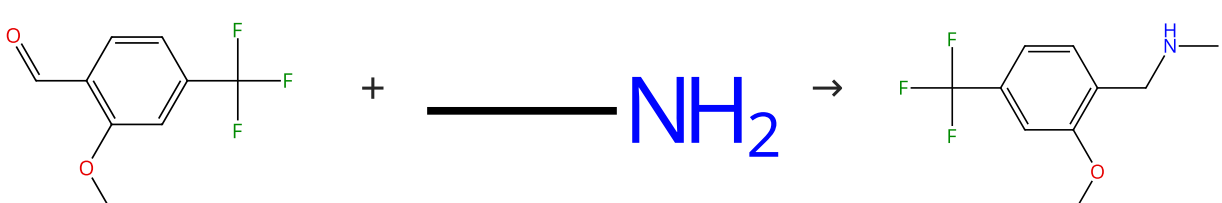
Suppliers (46)

Suppliers (42)

31-614-CAS-39391452	Steps: 1 Yield: 100%	Synthesis of novel benzylamine antimycotics and evaluation of their antimycotic potency
1.1 Reagents: Sodium triacetoxyborohydride Solvents: Tetrahydrofuran; 12 h, rt		By: Klimt, Monika; et al
1.2 Reagents: Sodium bicarbonate Solvents: Water		Archiv der Pharmazie (Weinheim, Germany) (2024), 357(5), 2300381.
1.3 Reagents: Hydrochloric acid Solvents: Diethyl ether		
Experimental Protocols		

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



The reaction scheme shows the conversion of 4-(4-methoxyphenyl)-2,2,2-trifluoroacetaldehyde to 4-(4-methoxyphenyl)-2,2,2-trifluoro-N-methylmethanamine. The starting material is a benzene ring with a methoxy group (-OCH₃) at the 1-position, a trifluoromethyl group (-CF₃) at the 4-position, and a formyl group (-CHO) at the 1-position. It reacts with an amine (NH₂) to form the product, which is a benzene ring with a methoxy group (-OCH₃) at the 1-position, a trifluoromethyl group (-CF₃) at the 4-position, and an N-methylmethanamine group (-CH₂NHCH₃) at the 1-position.

 Suppliers (66)

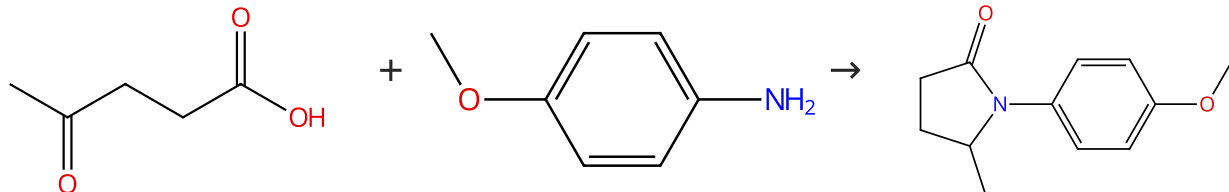
 Suppliers (115)

 Suppliers (7)

31-614-CAS-40490482	Steps: 1 Yield: 100%	Selective Defluorination of Trifluoromethyl Substituents by Conformationally Induced Remote Substitution
1.1 Solvents: Ethanol, Methanol; 1.5 h, rt; rt → 0 °C		By: Jesani, Mehul H.; et al
1.2 Reagents: Sodium borohydride; 0 °C; 0 °C → rt; 20 h, rt		Angewandte Chemie, International Edition (2024), 63(24), e202403477.
1.3 Reagents: Sodium hydroxide Solvents: Water		
Experimental Protocols		

Scheme 14 (2 Reactions)

Steps: 1 Yield: 100%



The reaction scheme shows the synthesis of 1-(4-methoxyphenyl)-2-methylpyrrolidin-3-one from 4-oxopentanoic acid and 4-methoxyaniline. The reactants are 4-oxopentanoic acid (a five-carbon chain with a ketone at C4 and a carboxylic acid at C1) and 4-methoxyaniline (a benzene ring with a methoxy group and an amino group in para positions). The product is a pyrrolidine ring with a methyl group at C2, a carbonyl at C3, and a 4-methoxyphenyl group at C1. The reaction is indicated by a plus sign and a right-pointing arrow.

Suppliers (98)

Suppliers (83)

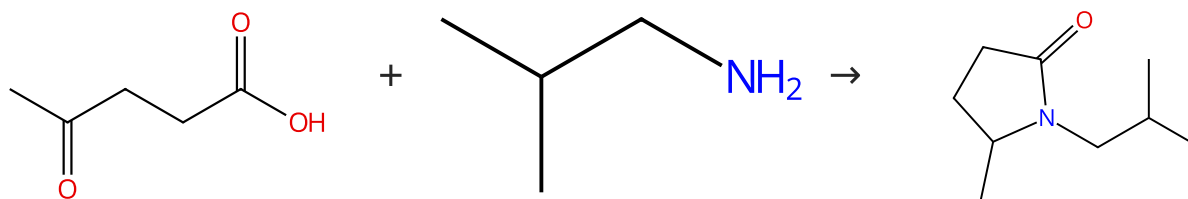
Suppliers (3)

31-614-CAS-42263159	Steps: 1 Yield: 100%	Sustainable Microwave-Assisted Routes to N-Substituted Pyrrolidones Using Natural Deep Eutectic Solvents as Non-innocent Solvents
1.1 Reagents: Triethoxysilane Solvents: 2242414-51-1; rt → 80 °C; 10 min, 5 bar, 80 °C		By: Tabasso, Silvia; et al
Experimental Protocols		ACS Sustainable Chemistry & Engineering (2024), 12(37), 13810-13815.

31-614-CAS-38832397	Steps: 1 Yield: 100%	Batch and Flow Green Microwave-Assisted Catalytic Conversion Of Levulinic Acid to Pyrrolidones
1.1 Reagents: Hydrogen Catalysts: Palladium, Gold (titania-doped highly mesoporous silica supported); 4 h, 20 bar, 130 °C		By: Calcio Gaudino, Emanuela; et al
Experimental Protocols		ChemSusChem (2024), 17(15), e202301200.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (60)

Suppliers (5)

31-614-CAS-42263160

Steps: 1 Yield: 100%

Sustainable Microwave-Assisted Routes to N-Substituted Pyrrolidones Using Natural Deep Eutectic Solvents as Non-innocent Solvents

By: Tabasso, Silvia; et al

ACS Sustainable Chemistry & Engineering (2024), 12(37), 13810-13815.

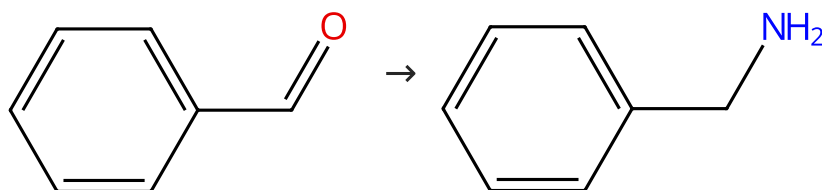
1.1 Reagents: Triethoxysilane

Solvents: 2242414-51-1; rt → 80 °C; 10 min, 5 bar, 80 °C

Experimental Protocols

Scheme 16 (2 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (69)

Suppliers (81)

31-614-CAS-38964579

Steps: 1 Yield: 100%

Highly active Co₁-N₃-O₁ single-atom catalyst for boosting catalytic synthesis of unsaturated benzylamines and arylamines

By: Han, Song; et al

Chemical Engineering Journal (Amsterdam, Netherlands) (2024), 480, 148330.

1.1 Catalysts: Cobalt (supported on hollow N-doped carbon nanospheres)

Solvents: Methanol; 10 min, rt

1.2 Reagents: Hydrogen, Ammonia; 3 h, 4 bar, 60 °C

Experimental Protocols

31-614-CAS-38860345

Steps: 1 Yield: 99%

Direct Synthesis of N-formamides by Integrating Reductive Amination of Ketones and Aldehydes with CO₂ Fixation in a Metal-Organic Framework

By: Huang, Wenyan; et al

Chemistry - A European Journal (2024), 30(7), e202303289.

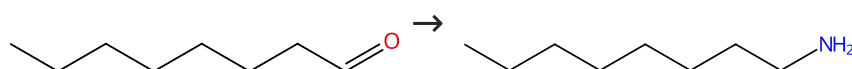
1.1 Reagents: Hydrogen, Ammonia

Catalysts: Ruthenium trichloride (compound with reaction product of H₄BPTC with CrCl₃)

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



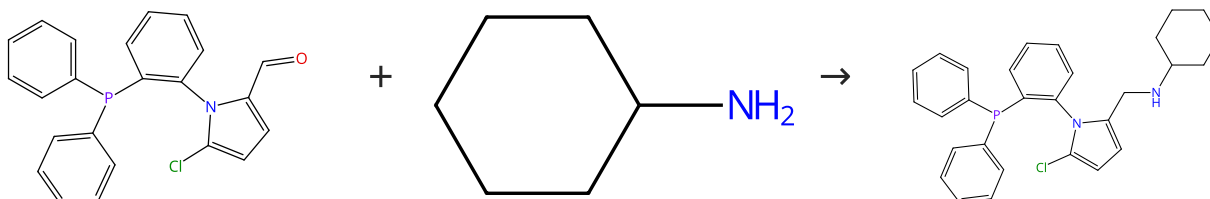
Suppliers (82)

Suppliers (79)

31-614-CAS-40238609	Steps: 1 Yield: 100%	Aliphatic amines from waste polyolefins by tandem pyrolysis, hydroformylation, and reductive amination
1.1 Reagents: Hydrogen, Ammonia Catalysts: Ruthenium Solvents: Methanol; 150 min, 40 bar, 90 °C		By: Li, Houqian; et al
Experimental Protocols		Green Chemistry (2024), 26(15), 8718-8727.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%

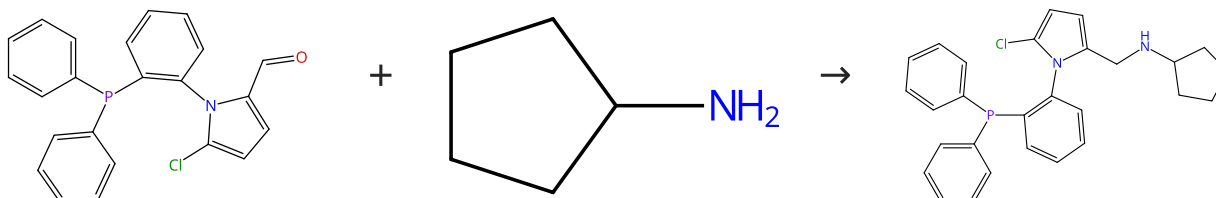


Suppliers (85)

31-614-CAS-41470173	Steps: 1 Yield: 99%	Catalytic Atroposelective Synthesis of C-N Axially Chiral Aminophosphines via Dynamic Kinetic Resolution
1.1 Solvents: Dichloromethane; 18 h, 60 °C; 60 °C → rt 1.2 Reagents: Formic acid, Triethylamine Catalysts: [<i>N</i> -[(1 <i>S</i> ,2 <i>S</i>)-2-(Amino-κ <i>N</i>)-1,2-diphenylethyl]-4-methylbenzenesulfonamidato-κ <i>M</i>]chloro[(1,2,3,4,5,6-η)-1-methyl-4-(1-methylethyl)benzene]ruthenium Solvents: Dichloromethane; 5 - 40 h, rt		By: Rodriguez-Franco, Carlos; et al
Experimental Protocols		Angewandte Chemie, International Edition (2024), 63(37), e202409524.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%

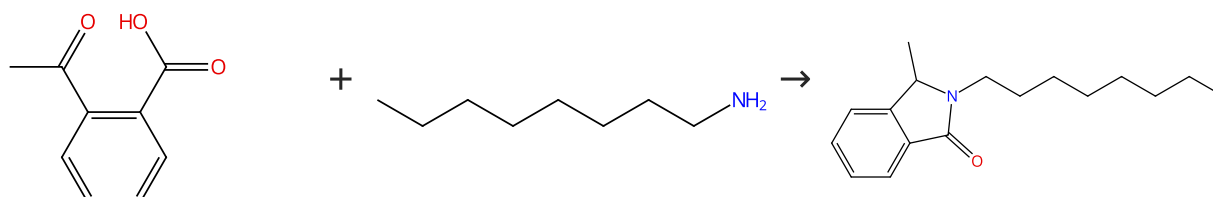


Suppliers (63)

31-614-CAS-41470185	Steps: 1 Yield: 99%	Catalytic Atroposelective Synthesis of C-N Axially Chiral Aminophosphines via Dynamic Kinetic Resolution
1.1 Solvents: Dichloromethane; 18 h, 60 °C; 60 °C → rt 1.2 Reagents: Formic acid, Triethylamine Catalysts: [<i>N</i> -[(1 <i>S</i> ,2 <i>S</i>)-2-(Amino-κ <i>N</i>)-1,2-diphenylethyl]-4-methylbenzenesulfonamidato-κ <i>M</i>]chloro[(1,2,3,4,5,6-η)-1-methyl-4-(1-methylethyl)benzene]ruthenium Solvents: Dichloromethane; 5 - 40 h, rt		By: Rodriguez-Franco, Carlos; et al
Experimental Protocols		Angewandte Chemie, International Edition (2024), 63(37), e202409524.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



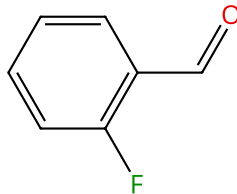
Suppliers (101)

Suppliers (79)

31-614-CAS-42058938	Steps: 1 Yield: 99%	A Homogeneous Nickel Catalyst for Reductive Amination of γ-Keto Acids using Hydrogen
1.1 Catalysts: Borate(1-), tetrafluoro-, nickel(2+) (2:1), hexahydrate, Triphos Solvents: 2,2,2-Trifluoroethanol; 10 - 15 min, rt		By: Chakraborty, Soumyadeep; et al
1.2 5 - 10 min, rt		ChemCatChem (2024), 16(22), e202400987.
1.3 Reagents: Hydrogen; 20 h, 30 bar, 110 °C		
Experimental Protocols		

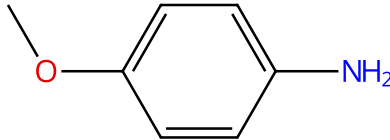
Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



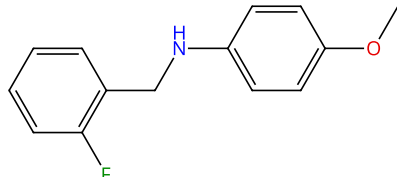
Suppliers (82)

+



Suppliers (83)

→

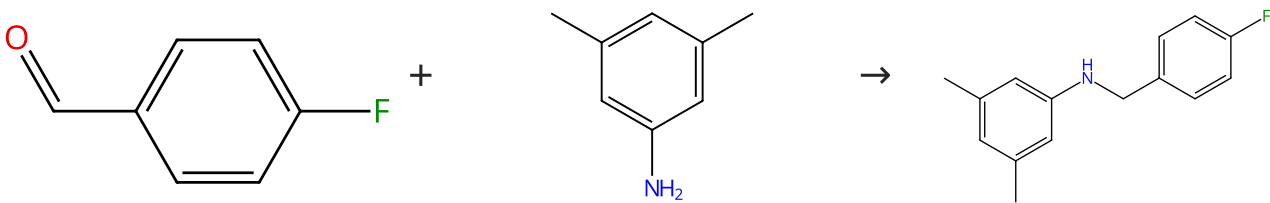


Suppliers (18)

31-614-CAS-38715683	Steps: 1 Yield: 99%	Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes
1.1 Reagents: Hydrogen Catalysts: Palladium, Chitosan Solvents: Toluene; 12 h, 3 MPa, rt $\rightarrow$ 120 °C		By: Yin, Xiaogang; et al
Experimental Protocols		Applied Surface Science (2024), 648, 159047.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



The reaction scheme shows the synthesis of N-(4-fluorophenyl)-3,5-dimethylbenzylamine. The reactants are 4-fluorobenzaldehyde (a benzene ring with a formyl group at position 1 and a fluorine atom at position 4) and 3,5-dimethylaniline (a benzene ring with an amino group at position 1 and methyl groups at positions 3 and 5). The product is N-(4-fluorophenyl)-3,5-dimethylbenzylamine, where the aldehyde group of 4-fluorobenzaldehyde has been converted to a secondary amine linkage (-NH-) connected to the 3,5-dimethylphenyl group.

Suppliers (96)

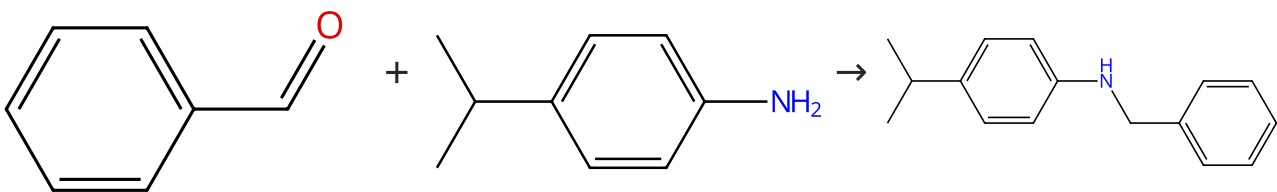
Suppliers (63)

Suppliers (11)

31-614-CAS-38715651	Steps: 1 Yield: 99%	Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes
1.1 Reagents: Hydrogen Catalysts: Palladium, Chitosan Solvents: Toluene; 12 h, 3 MPa, rt $\rightarrow$ 120 °C		By: Yin, Xiaogang; et al
Experimental Protocols		Applied Surface Science (2024), 648, 159047.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



The reaction scheme shows the reductive amination of benzaldehyde with 4-isopropylaniline. Benzaldehyde (a benzene ring with an aldehyde group, -CHO) reacts with 4-isopropylaniline (a benzene ring with an isopropyl group and an amino group, -NH₂, in para positions). The product is N-benzyl-4-isopropylaniline, where the amino group of the starting material has been converted to a secondary amine, -NH-CH₂-C₆H₅.

 Suppliers (69)

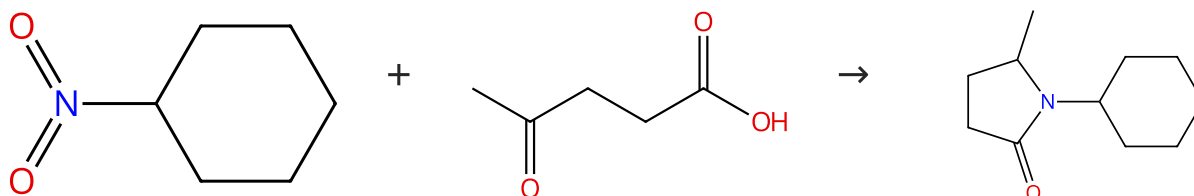
 Suppliers (81)

 Suppliers (11)

31-614-CAS-38944930	Steps: 1 Yield: 99%	A highly active and chemoselective homobimetallic ruthenium catalyst for one-pot reductive amination in water
1.1 Solvents: Water; 5 min, rt		By: Deshmukh, Gopal; et al
1.2 Reagents: Formic acid, Sodium formate Catalysts: 3026432-81-2; 4 h, pH 3.5, 80 °C		Green Chemistry (2024), 26(3), 1610-1626.
Experimental Protocols		

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (55)

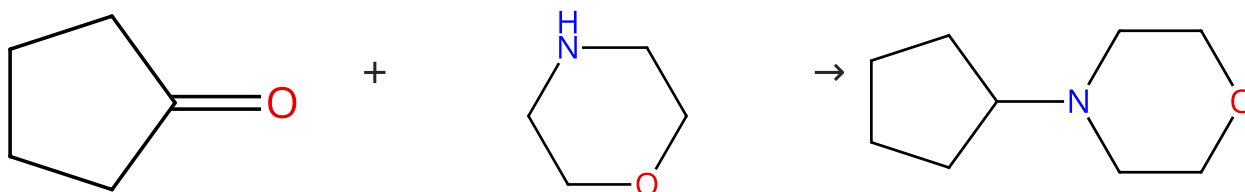
Suppliers (98)

Suppliers (3)

31-614-CAS-42832544	Steps: 1 Yield: 99%	N, P co-doped porous biochar supported electron deficient Ni^{δ+} for domino reductive amination and amidation of levulinic acid with nitroarenes to N-substituted pyrrolidones using methyl formate
1.1 Reagents: Methyl formate Catalysts: Nickel phosphide (Ni ₂ P) (N, P doped, biochar supported) Solvents: Methanol, Water; 16 h, rt → 120 °C		By: Gong, Honghui; et al
Experimental Protocols		Journal of Catalysis (2024), 440, 115848.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (78)

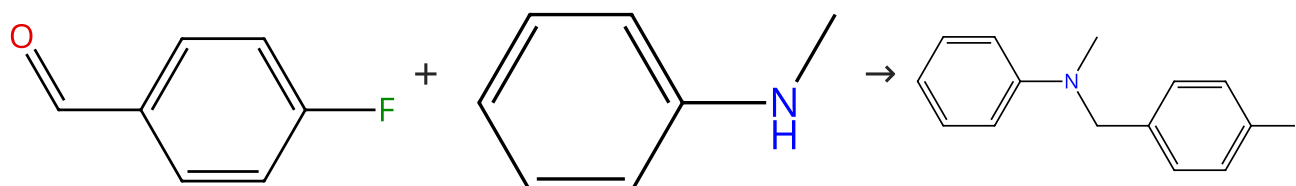
Suppliers (88)

Suppliers (48)

31-614-CAS-43947244	Steps: 1 Yield: 99%	Encapsulation of Pt species into MFI zeolite with tunable acid sites boosts reductive amination towards tertiary amines
1.1 Reagents: Hydrogen Catalysts: Platinum Solvents: Ethanol; 2 h, 15 bar, 80 °C		By: Xiong, Zhuo; et al
Experimental Protocols		AIChE Journal (2025), 71(4), e18710.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (96)

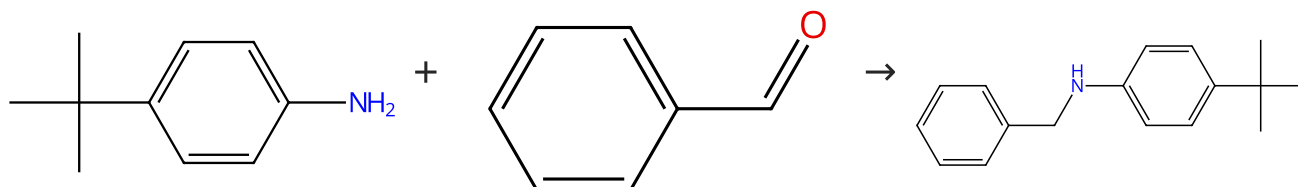
Suppliers (73)

Suppliers (8)

31-614-CAS-38715686	Steps: 1 Yield: 99%	Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes By: Yin, Xiaogang; et al Applied Surface Science (2024), 648, 159047.
1.1 Reagents: Hydrogen Catalysts: Palladium, Chitosan Solvents: Toluene; 8 h, 3 MPa, rt → 120 °C		
Experimental Protocols		

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

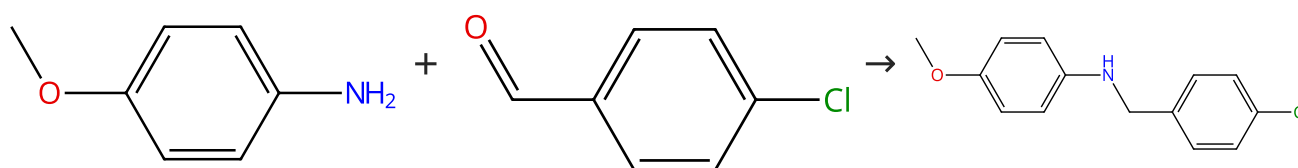
Suppliers (69)

Suppliers (16)

31-614-CAS-38944932	Steps: 1 Yield: 99%	A highly active and chemoselective homobimetallic ruthenium catalyst for one-pot reductive amination in water By: Deshmukh, Gopal; et al Green Chemistry (2024), 26(3), 1610-1626.
1.1 Solvents: Water; 5 min, rt 1.2 Reagents: Formic acid, Sodium formate Catalysts: 3026432-81-2; 4 h, pH 3.5, 80 °C		
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

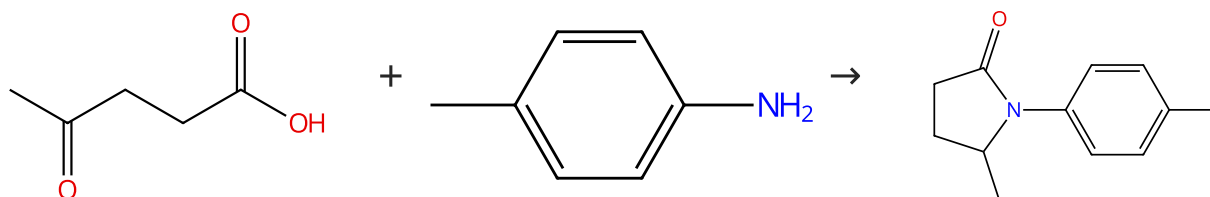
Suppliers (99)

Suppliers (41)

31-614-CAS-38715657	Steps: 1 Yield: 99%	Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes By: Yin, Xiaogang; et al Applied Surface Science (2024), 648, 159047.
1.1 Reagents: Hydrogen Catalysts: Palladium, Chitosan Solvents: Toluene; 12 h, 3 MPa, rt → 120 °C		
Experimental Protocols		

Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

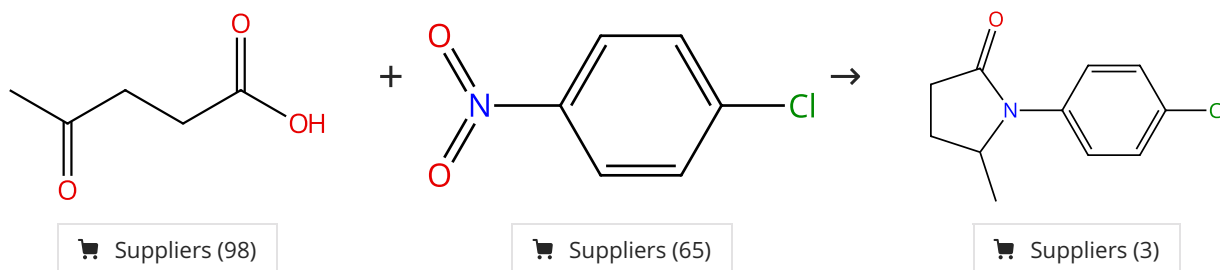
Suppliers (76)

Suppliers (2)

31-614-CAS-38832406	Steps: 1 Yield: 99%	Batch and Flow Green Microwave-Assisted Catalytic Conversion Of Levulinic Acid to Pyrrolidones
1.1 Reagents: Hydrogen Catalysts: Palladium, Gold (titania-doped highly mesoporous silica supported); 1.5 h, 5 bar, 150 °C		By: Calcio Gaudino, Emanuela; et al
Experimental Protocols		ChemSusChem (2024), 17(15), e202301200.

Scheme 30 (2 Reactions)

Steps: 1 Yield: 99%

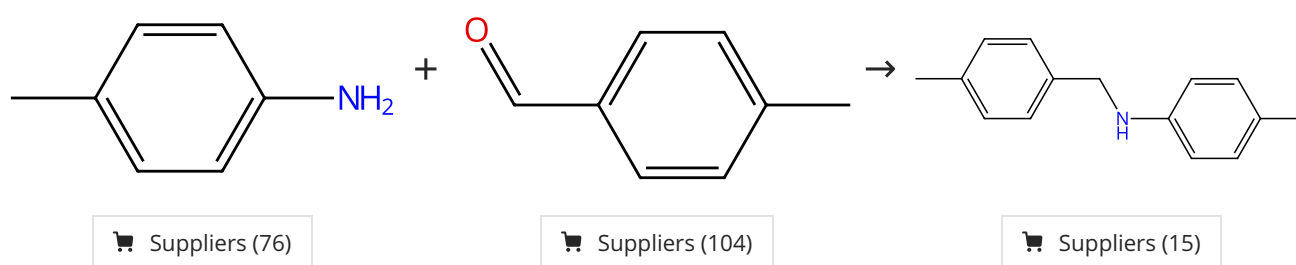


31-614-CAS-38968287	Steps: 1 Yield: 99%	Room temperature synthesis of N-substituted pyrrolidone from levulinic acid and nitrobenzene via a tandem hydrogenation and reductive amination process
1.1 Reagents: Hydrogen Catalysts: Niobium pentoxide, Platinum Solvents: Methanol; 24 h, 30 °C		By: Li, Shiting; et al
Experimental Protocols		Journal of Catalysis (2024), 429, 115191.

31-614-CAS-42832518	Steps: 1 Yield: 99%	N, P co-doped porous biochar supported electron deficient Ni^{δ+} for domino reductive amination and amidation of levulinic acid with nitroarenes to N-substituted pyrrolidones using methyl formate
1.1 Reagents: Methyl formate Catalysts: Nickel phosphide (Ni ₂ P) (N, P doped, biochar supported) Solvents: Methanol, Water; 16 h, rt → 120 °C		By: Gong, Honghui; et al
Experimental Protocols		Journal of Catalysis (2024), 440, 115848.

Scheme 31 (1 Reaction)

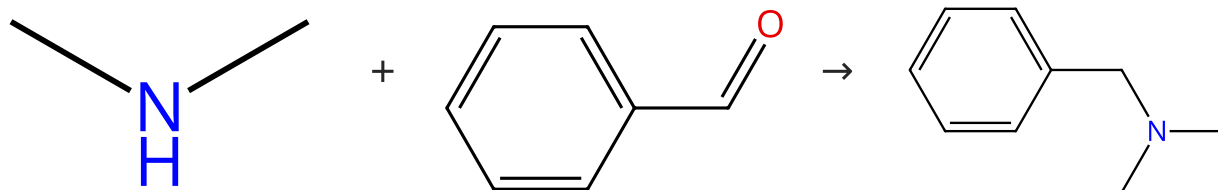
Steps: 1 Yield: 99%



31-614-CAS-38715674	Steps: 1 Yield: 99%	Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes
1.1 Reagents: Hydrogen Catalysts: Palladium, Chitosan Solvents: Toluene; 12 h, 3 MPa, rt → 120 °C		By: Yin, Xiaogang; et al
Experimental Protocols		Applied Surface Science (2024), 648, 159047.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (118)

Suppliers (69)

Suppliers (65)

31-614-CAS-43947216

Steps: 1 Yield: 99%

Encapsulation of Pt species into MFI zeolite with tunable acid sites boosts reductive amination towards tertiary amines

By: Xiong, Zhuo; et al

AIChE Journal (2025), 71(4), e18710.

1.1 Reagents: Hydrogen

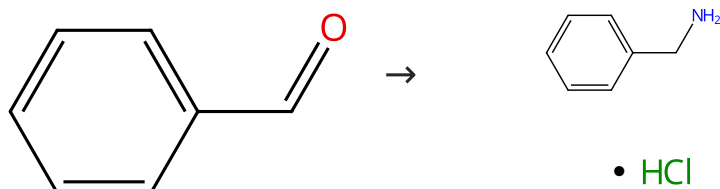
Catalysts: Platinum

Solvents: Ethanol; 2 h, 15 bar, 80 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (69)

Suppliers (82)

31-614-CAS-42278658

Steps: 1 Yield: 99%

Reductive Amination of Aldehyde and Ketone with Ammonia and H₂ by an In Situ-Generated Cobalt Catalyst under Mild Conditions

By: Zhang, Shiyun; et al

Organic Letters (2024), 26(34), 7122-7127.

1.1 Catalysts: Cobalt chloride (CoCl₂), Sodium borohydride

Solvents: 1,4-Dioxane; 3 h, rt

1.2 Reagents: Hydrogen, Ammonia

Solvents: 1,4-Dioxane; 12 h, 1 bar, 80 °C; 80 °C → rt

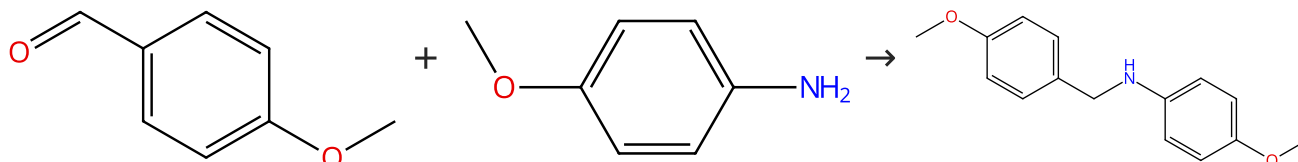
1.3 Reagents: Hydrochloric acid

Solvents: Diethyl ether, Ethyl acetate; 4 - 5 h, rt

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (128)

Suppliers (83)

Suppliers (42)

31-614-CAS-38715652

Steps: 1 Yield: 99%

Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes

By: Yin, Xiaogang; et al

Applied Surface Science (2024), 648, 159047.

1.1 Reagents: Hydrogen

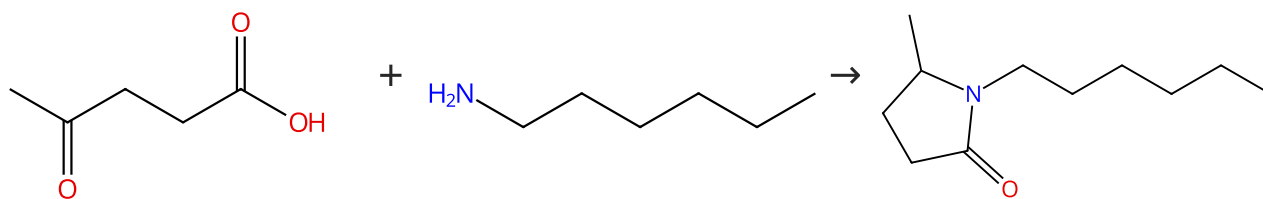
Catalysts: Palladium, Chitosan

Solvents: Toluene; 12 h, 3 MPa, rt → 120 °C

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (71)

Suppliers (6)

31-614-CAS-42058928

Steps: 1 Yield: 99%

A Homogeneous Nickel Catalyst for Reductive Amination of γ -Keto Acids using Hydrogen

By: Chakraborty, Soumyadeep; et al

ChemCatChem (2024), 16(22), e202400987.

1.1 **Catalysts:** Borate(1-), tetrafluoro-, nickel(2+) (2:1), hexahydrate, Triphos**Solvents:** 2,2,2-Trifluoroethanol; 10 - 15 min, rt

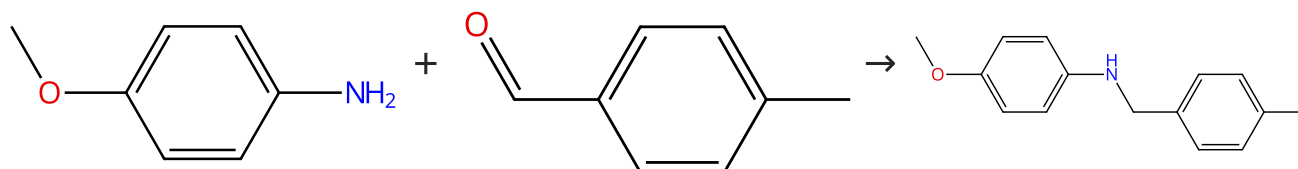
1.2 5 - 10 min, rt

1.3 **Reagents:** Hydrogen; 20 h, 30 bar, 110 °C

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (104)

Suppliers (13)

31-614-CAS-38715664

Steps: 1 Yield: 99%

Highly dispersed and stable Pd nano-catalyst loaded on chitosan nanoporous carbon microspheres for one-pot reductive amination of amines with aldehydes

By: Yin, Xiaogang; et al

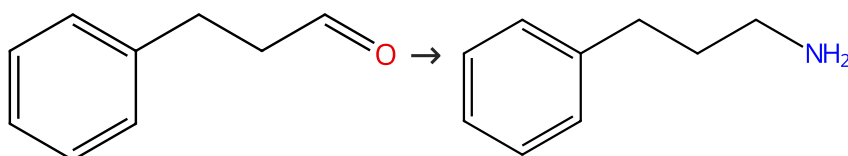
Applied Surface Science (2024), 648, 159047.

1.1 **Reagents:** Hydrogen**Catalysts:** Palladium, Chitosan**Solvents:** Toluene; 12 h, 3 MPa, rt $\rightarrow$ 120 °C

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (76)

31-614-CAS-40309978

Steps: 1 Yield: 99%

Efficient reductive amination of furfural to furfurylamine promoted synergistically by surface Co^0 and oxygen-vacant CoO_x

By: Wei, Yilin; et al

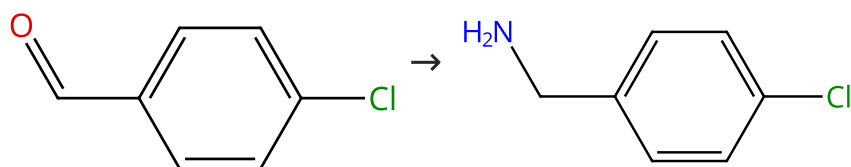
Fuel (2024), 369, 131703.

1.1 **Reagents:** Hydrogen, Hydrazine hydrate (1:1)**Catalysts:** Cobalt oxide**Solvents:** Methanol; 5 h, 3 MPa, 60 °C

Experimental Protocols

Scheme 38 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (99)

Suppliers (87)

31-614-CAS-40309975

Steps: 1 Yield: 99%

Efficient reductive amination of furfural to furfurylamine promoted synergistically by surface Co^0 and oxygen-vacant CoO_x

By: Wei, Yilin; et al

Fuel (2024), 369, 131703.

1.1 Reagents: Hydrogen, Hydrazine hydrate (1:1)

Catalysts: Cobalt oxide

Solvents: Methanol; 4 h, 3 MPa, 60 °C

Experimental Protocols

31-614-CAS-38964591

Steps: 1 Yield: 99%

Highly active $\text{Co}_1\text{-N}_3\text{-O}_1$ single-atom catalyst for boosting catalytic synthesis of unsaturated benzylamines and arylamines

By: Han, Song; et al

Chemical Engineering Journal (Amsterdam, Netherlands) (2024), 480, 148330.

1.1 Catalysts: Cobalt (supported on hollow N-doped carbon nanospheres)

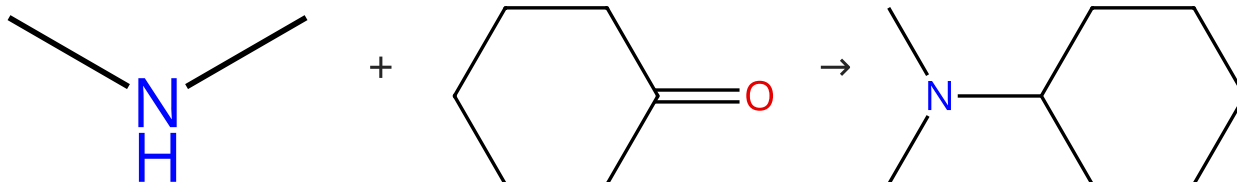
Solvents: Methanol; 10 min, rt

1.2 Reagents: Hydrogen, Ammonia; 4 h, 4 bar, 90 °C

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (118)

Suppliers (102)

Suppliers (59)

31-614-CAS-41287293

Steps: 1 Yield: 99%

Unveiling synergistic catalysis in reductive amination through a $\text{UiO-66-NH}_2@(\text{PdAu})@(\text{MOF-808})$ sandwich nanostructure

By: Han, Tianchen; et al

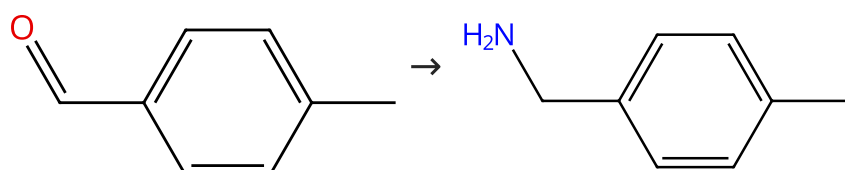
Applied Catalysis, A: General (2024), 685, 119884.

1.1 Reagents: Hydrogen
Catalysts: Gold compd. with palladium (1: 1), Hexakis[μ -[2-amino-1,4-benzenedicarboxylato(2-)- $\kappa\text{O}^1:\kappa\text{O}^1$]]tetra- μ_3 -hydroxytetra- μ_3 -oxohexazirconium, Bis[μ -[1,3,5-benzene tricarboxylato(3-)- $\kappa\text{O}^1:\kappa\text{O}^1$]]hexakis[μ -(formato- $\kappa\text{O}:\kappa\text{O}$)]tetra- μ_3 -hydroxytetra- μ_3 -oxohexazirconium
Solvents: Methanol; 6 h, 1 atm, 30 °C

Experimental Protocols

Scheme 40 (3 Reactions)

Steps: 1 Yield: 99%



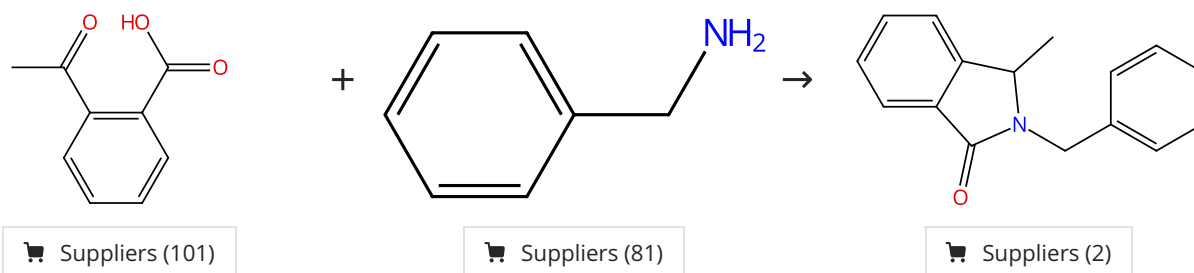
Suppliers (104)

Suppliers (80)

31-614-CAS-44209161 Steps: 1 Yield: 99% 1.1 Reagents: Ammonia Solvents: Methanol; 1 h, 35 °C 1.2 Reagents: Hydrogen Catalysts: Alumina, Nickel, Cesium; 3.5 min, 1 MPa, 50 °C Experimental Protocols	Enhanced efficiency and stability of Cs-modified Ni-based catalysts for selective synthesis of primary amines By: Zhang, Jiahao; et al Chemical Engineering Journal (Amsterdam, Netherlands) (2025), 503, 158280.
31-614-CAS-38964586 Steps: 1 Yield: 99% 1.1 Catalysts: Cobalt (supported on hollow N-doped carbon nanospheres) Solvents: Methanol; 10 min, rt 1.2 Reagents: Hydrogen, Ammonia; 4 h, 4 bar, 60 °C Experimental Protocols	Highly active Co₁-N₃-O₁ single-atom catalyst for boosting catalytic synthesis of unsaturated benzylamines and arylamines By: Han, Song; et al Chemical Engineering Journal (Amsterdam, Netherlands) (2024), 480, 148330.
31-614-CAS-41150397 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Ammonia Catalysts: Ruthenium Solvents: Methanol; 2 h, 2 MPa, 50 °C Experimental Protocols	Electron-Rich Ru Supported on N-Doped Coffee Biochar for Selective Reductive Amination of Furfural to Furfurylamine By: Gong, Honghui; et al Langmuir (2024), 40(17), 8950-8960.

Scheme 41 (1 Reaction)

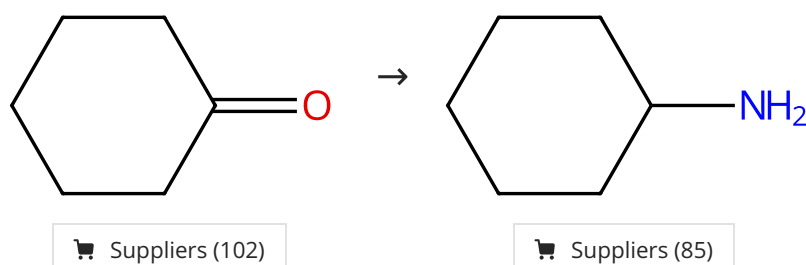
Steps: 1 Yield: 99%



31-614-CAS-42058940 Steps: 1 Yield: 99% 1.1 Catalysts: Borate(1-), tetrafluoro-, nickel(2+) (2:1), hexahydrate, Triphos Solvents: 2,2,2-Trifluoroethanol; 10 - 15 min, rt 1.2 5 - 10 min, rt 1.3 Reagents: Hydrogen; 20 h, 30 bar, 110 °C Experimental Protocols	A Homogeneous Nickel Catalyst for Reductive Amination of γ-Keto Acids using Hydrogen By: Chakraborty, Soumyadeep; et al ChemCatChem (2024), 16(22), e202400987.
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Scheme 42 (2 Reactions)

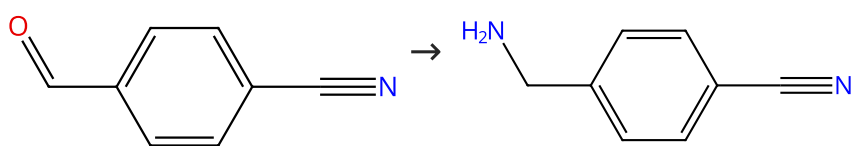
Steps: 1 Yield: 99%



31-614-CAS-44343073 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Ammonia Catalysts: Platinum, Titania Solvents: Methanol; 5 h, 100 °C Experimental Protocols	Efficient Reductive Amination of Furfural to a Primary Amine on a Pt/TiO₂ Catalyst: A Manifestation of the Nanocluster Proximity Effect By: Du, Xiaorui; et al ACS Catalysis (2025), 15(6), 4654-4664.
31-614-CAS-40309974 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Hydrazine hydrate (1:1) Catalysts: Cobalt oxide Solvents: Methanol; 4 h, 3 MPa, 60 °C Experimental Protocols	Efficient reductive amination of furfural to furfurylamine promoted synergistically by surface Co⁰ and oxygen-vacant CoO_x By: Wei, Yilin; et al Fuel (2024), 369, 131703.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (70)

31-614-CAS-42143052 Steps: 1 Yield: 99% 1.1 Reagents: Hydrogen, Ammonia Catalysts: Zirconium dioxide, Ruthenium Solvents: Methanol; 3 h, 20 bar, 80 °C Experimental Protocols	Surface Acidic Species-Driven Reductive Amination of Furfural with Ru/T-ZrO₂ By: Saini, Kanika; et al ChemSusChem (2025), 18(2), e202401277.
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